

A Decade of Malaria Prevalence in Pankshin LGA, Plateau State, Nigeria: Trends, Demographics, and Reporting Consistency towards Stronger Rural Prevention Measures

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ABSTRACT

*Malaria remains a major public health challenge in Nigeria, with rural communities bearing the greatest burden. This study analyzed malaria prevalence in Pankshin LGA, Plateau State, Nigeria, over a decade (2014–2024), focusing on temporal trends, seasonal variation, demographic disparities, gender differences, and data quality. Retrospective data were obtained from monthly outpatient registers of seven randomly selected primary health clinics using a researcher-designed checklist. Statistical analyses, including Pearson correlation, linear regression, ANOVA with Tukey HSD tests, and Chi-squared tests, were conducted in R software. Results revealed a significant increase in malaria cases, rising from 92 in 2014 to 1104 in 2024 ($r = 0.776$, $p = 0.005$). Monthly cases peaked visually in August, **although** no consistent seasonal pattern was detected ($F(10, 449) = 1.15$, $p = 0.0323$). Significant disparities were observed across age groups ($F(4, 2495) = 44.86$, $p < 0.001$), with school-aged children (6–16 years) and young adults (17–30 years) being the most affected. **Demographic variation was also observed ($F(3, 1996) = 12.7$, $p < 0.001$), with a disproportionate impact on pregnant women, students, and farmers.** Females recorded more cases (5134) than males (3803). Clinic-level heterogeneity was marked ($F(7, 492) = 278.9$, $p < 0.001$), and data inconsistencies included a complete absence of 2018 records. The study highlights a rising malaria burden and the urgent need for targeted interventions, including strengthened antenatal care, school-based programs, and vector control in farming communities. Robust, resilient surveillance systems are critical for accurate monitoring and progress toward malaria elimination.*

Keywords: Malaria, Pankshin, **Socio-economy**, Plateau state, Malaria in children

INTRODUCTION

Malaria continues to pose a major public health challenge in Nigeria, contributing significantly to the global burden. In 2022, Nigeria recorded an estimated 67 million malaria cases and 200,000 deaths, accounting for about 27% of the global total.¹¹ Vulnerable groups, including children under five years, pregnant women, and rural populations, are disproportionately affected, worsening

health inequalities.^{9,10,12} Transmission in Nigeria is perennial, with seasonal peaks influenced by rainfall patterns that favor mosquito breeding.¹¹ Prevalence varies across regions and occupations, with rural dwellers and farmers particularly exposed due to proximity to breeding sites.^{2,6}

Children under five account for a high share of global malaria deaths, largely due to underdeveloped immunity, poor housing, and high exposure to bites.^{3,10} Pregnant women face heightened risks linked to weakened immunity during pregnancy, leading to anemia, low birth weight, and miscarriages.^{9,12} Gender disparities have also been reported, with females showing higher prevalence, attributed to pregnancy vulnerability and health-seeking behaviors.^{1,12} Occupational

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exposure further elevates risk for farmers working outdoors near mosquito habitats.^{6,2}

Control measures in Nigeria include insecticide-treated nets (ITNs), indoor residual spraying (IRS), and artemisinin-based combination therapies (ACTs). Despite these, malaria remains high due to insecticide resistance, limited funding, and weak health systems.^{12,7,2} The National Malaria Strategic Plan (NMSP) 2021–2025 targets a reduction of parasite prevalence to <10% and mortality to <50 deaths per 1,000 by 2025, but achieving this requires locally generated evidence.⁴

Effective surveillance depends on reliable reporting, yet inconsistencies persist, with clinics sometimes failing to submit data, limiting accurate trend analysis and planning.^{5,6} Social factors such as poverty, rural residence, and low maternal education also exacerbate vulnerability among children.⁸

This study analyzes malaria prevalence in seven rural communities of Pankshin LGA, Plateau State, over a decade (2014–2024). It examines yearly and monthly patterns,^{11,6} age-group variations,¹⁰ demographic and gender disparities,¹ and reporting consistency.¹² The hypotheses tested are: (i) no significant increase in cases over time; (ii) uniform monthly distribution with no seasonal variation; (iii) equal prevalence across age groups; and (iv) no demographic disparities. Testing these provides insights for policy, surveillance, and targeted interventions in Pankshin and similar endemic settings.

MATERIALS AND METHOD

Study site

This study was conducted in Pankshin LGA, Plateau State, North-Central Nigeria. The area is largely agrarian, with subsistence farming increasing outdoor exposure to malaria vectors. Plateau's high-altitude climate features a wet season (April–October) that fosters *Anopheles* breeding and sustains transmission. Limited healthcare, poor infrastructure, and inadequate housing heighten vulnerability, making Pankshin a

critical site for malaria surveillance and control in rural North-Central Nigeria.

Data Collection

Malaria case records were retrospectively extracted from seven primary-health clinics in rural communities of Pankshin LGA, Plateau State, Nigeria. To minimize selection bias and ensure geographic spread, clinics were chosen using simple random sampling from the full list of government-registered facilities. Monthly outpatient registers from January 2014 to December 2024 were reviewed in situ. Variables transcribed into a spreadsheet included: year, month, total confirmed malaria cases, sex, five age groups (1–5, 6–16, 17–30, 31–50, ≥51 years), and six demographic/occupational categories (pregnant women, students, farmers, civil servants, traders, others). A double-entry process by two independent clerks ensured data integrity, with discrepancies reconciled against original registers. Ambiguous entries were coded as missing. Only aggregated counts were recorded, ensuring no personal identifiers and adherence to ethical standards of confidentiality.

Data Analysis

Analyses were performed using R 4.3.0 (R Core Team, 2024). Descriptive summaries used *dplyr* and *tidyr*, while *ggplot2* generated visualizations, including grouped bar charts and time-series plots with regression lines. Temporal trends were examined using Pearson correlation and simple linear regression to quantify yearly change and 95% confidence intervals. Seasonal variation was tested with one-way ANOVA across months; where significant, Tukey HSD identified pairwise differences. Age-group counts were reshaped into long format and analyzed by one-way ANOVA with Tukey HSD to locate specific disparities. Demographic/occupational categories (pregnant women, students, farmers, civil servants, traders, others) were tested similarly. Clinic-level heterogeneity was assessed using one-way ANOVA with Tukey HSD. Reporting consistency was

evaluated by categorizing clinics as “consistent” (≥ 8 years reporting) or “sporadic” (< 8 years). Separate regressions were fitted for each group, and an interaction model tested whether temporal trends differed by reporting pattern. All tests were

two-tailed with $\alpha = 0.05$. Graphics incorporated 95% confidence ribbons for intuitive interpretation of significance.

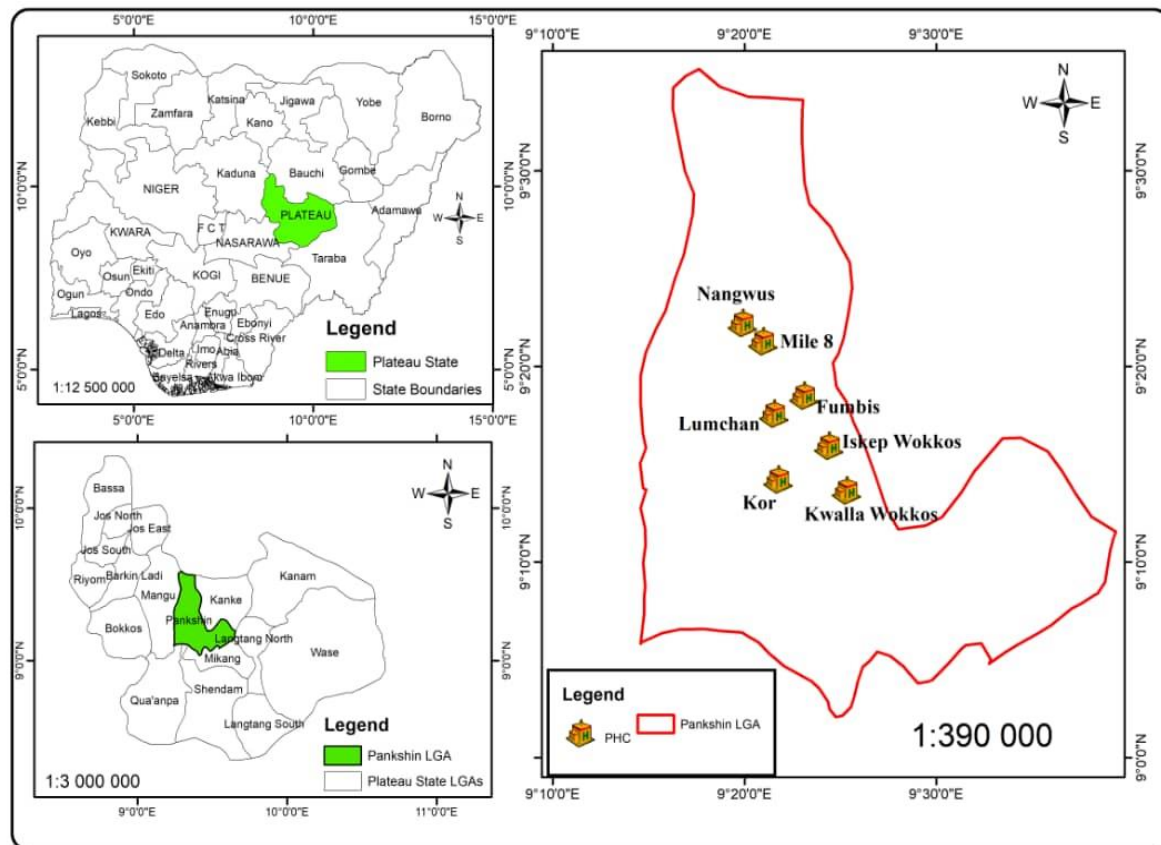


Fig. 1: Maps showing Locations of the Seven Primary Health Care Where We Collected our Data

RESULTS

Overall Trend (2014–2024)

Malaria cases rose markedly from 92 in 2014 to 1104 in 2024, despite fluctuations such as a dip in 2019–2020. A strong positive correlation was observed between year and total cases ($r = 0.776$, $p = 0.00495$). Linear regression confirmed a significant upward trend, with an average annual increase of 65 cases ($t = 3.696$, $p = 0.00495$). The model explained 60.3% of the variance ($R^2 = 0.603$; $F(1, 9) = 13.66$, $p = 0.00495$). This is seen in Figure 2.

Seasonal Variation

Monthly counts peaked in August (1010) and were lowest in March (527). However, ANOVA showed no significant seasonal effect ($F(10, 449) = 1.15$, $p = 0.323$) (Figure 3)

Age-Group Differences

The 6–16 years group recorded the highest cases (2935), followed by 17–30 (2593), 1–5 (2282), 31–50 (1487), and ≥ 51 (762). ANOVA confirmed significant differences ($F(4, 2495) = 44.86$, $p < 0.001$). Tukey tests showed younger groups (1–30 years) had significantly higher cases than older groups (31+) (Figure 4).

Demographic/Occupational Differences

Pregnant women (2296), students (2029), and farmers (1756) bore the highest burdens. ANOVA revealed significant disparities ($F(3, 1996) = 12.7, p < 0.001$).

The “Others” group had significantly lower cases than the other three (Figure 5).

Gender Differences

Females recorded more cases (5134) than males (3803).

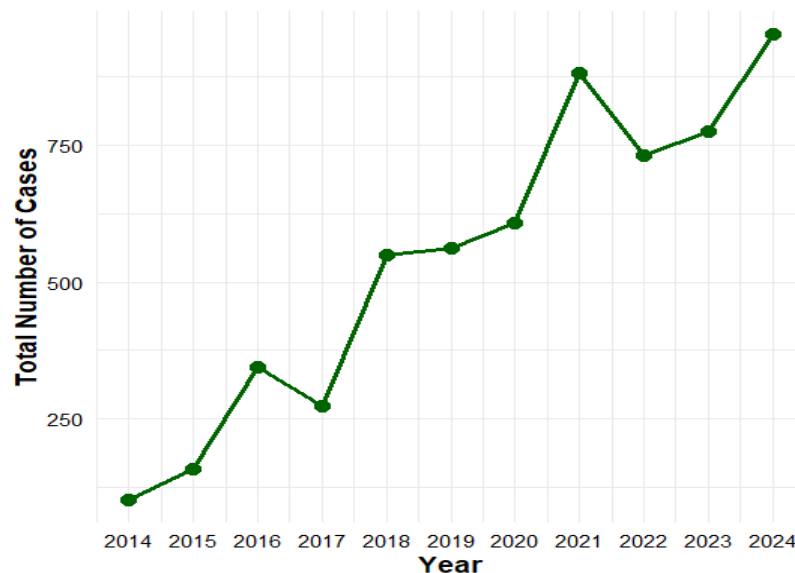


Fig. 2: The yearly trend of malaria cases from 2014 to 2024, derived from clinic data, reveals a significant increase in prevalence over the decade

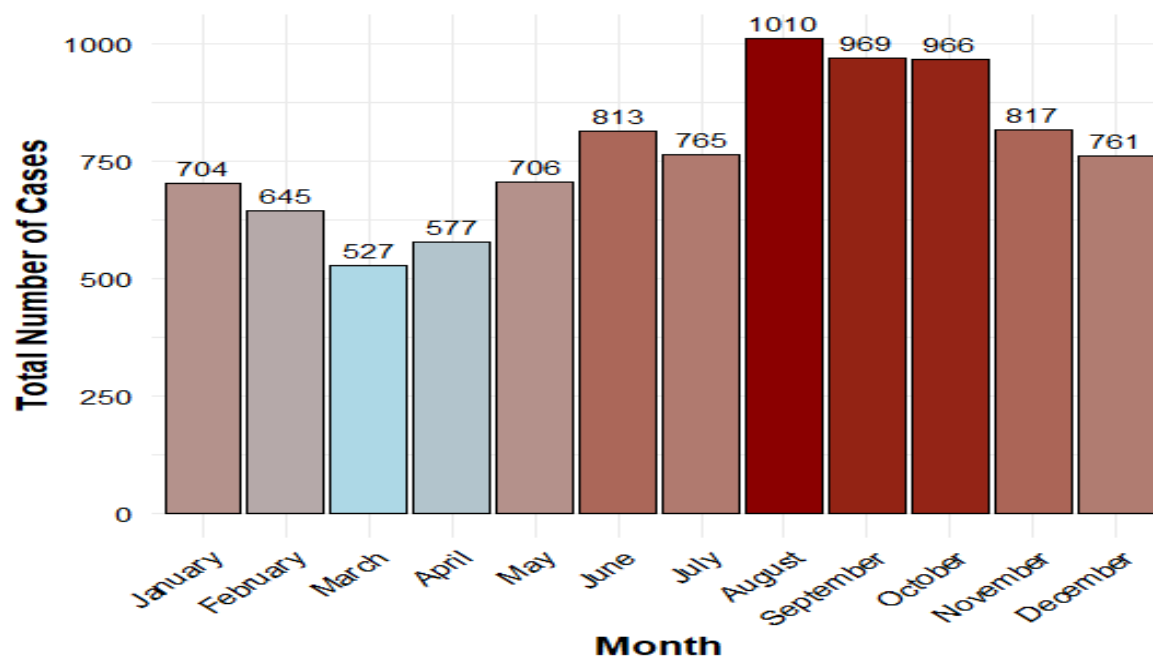


Fig.3: Bar charts Representing monthly distribution of malaria cases throughout the year, as derived from clinic data in Pankshin LGA, Plateau State, Nigeria exhibits significant variation with Rainy season Peaking

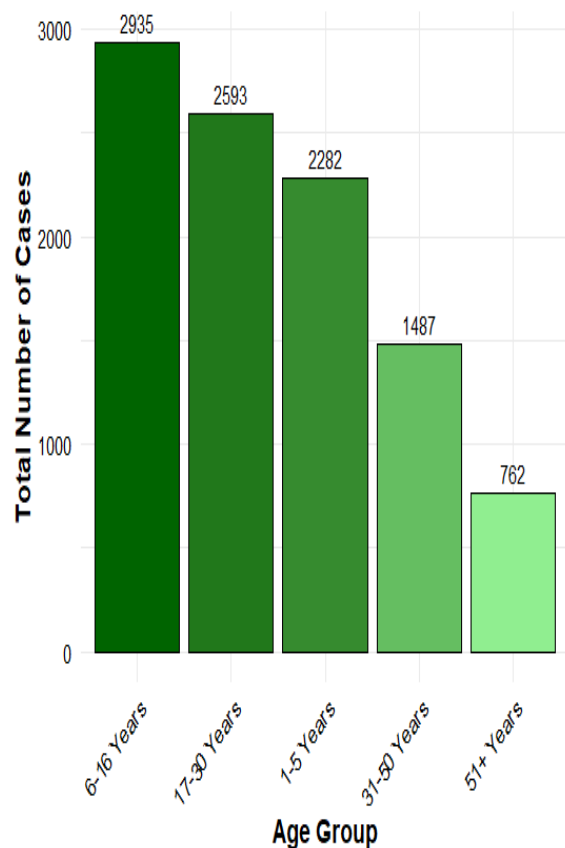


Fig. 4: Bar charts Showing total Number of Malaria Cases by age over 10 years period shows that children and adolescence are the most Affected

Clinic-Level Heterogeneity

Reporting varied widely, with a total absence in 2018. ANOVA showed significant clinic-level differences ($F(7, 492) = 278.9, p < 0.001$). Post-hoc tests revealed Mile 8 Fier and KOR reported significantly higher cases, while Butal Wookos and Kwalla had lower averages.

DISCUSSION

This study provides crucial insights into the evolving epidemiology of malaria in rural communities of Pankshin Local Government Area, Plateau State, Nigeria, by evaluating key trends and patterns against established epidemiological assumptions. Our findings confirm that malaria remains a significant public health challenge in the region, echoing the broader national and global context where Nigeria accounts for a

substantial proportion of the global malaria burden.¹¹

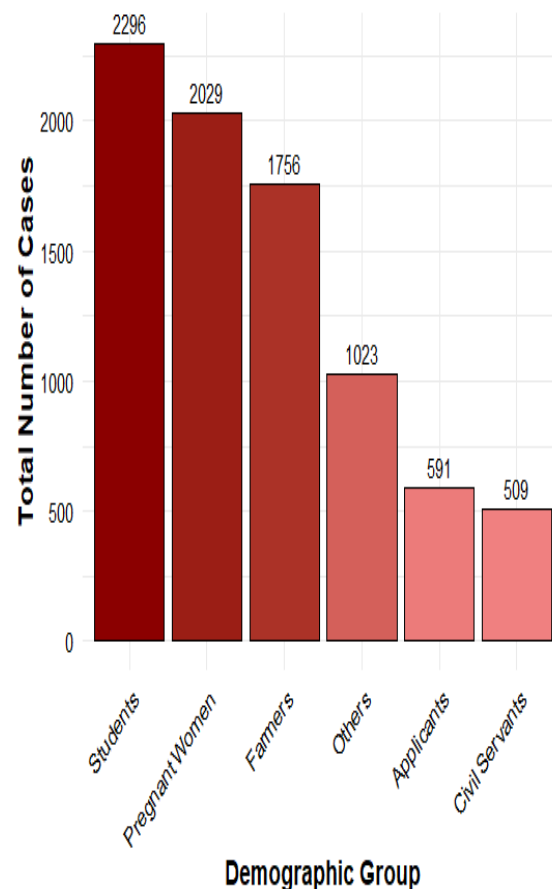


Fig. 5: Bar Charts Showing the Malaria Cases by Demographic Group in Pankshin LGA, Plateau State, Nigeria have Students, Pregnant women and Famers as the most Infected

The analysis of yearly trends (Figure 1) revealed a statistically significant increase in malaria cases from 2014 to 2024, contradicting our initial null hypothesis of constant case numbers. This upward trajectory aligns with the persistent high malaria burden reported across Nigeria despite ongoing control efforts.^{12,7} The observed rise could reflect genuine increases in transmission, potentially driven by factors such as environmental changes, which can create more favorable conditions for mosquito breeding, or challenges in the consistent application of vector control measures in these rural settings. Furthermore, it is plausible that improved

data collection efforts and increased awareness over the decade, particularly following the recovery of reporting consistency from 2020 onwards, may have contributed to the higher case numbers recorded in later years. This implies that earlier years might have experienced underreporting, making the true increase in actual cases potentially less steep than the raw figures suggest, while simultaneously indicating a strengthening of surveillance capabilities.

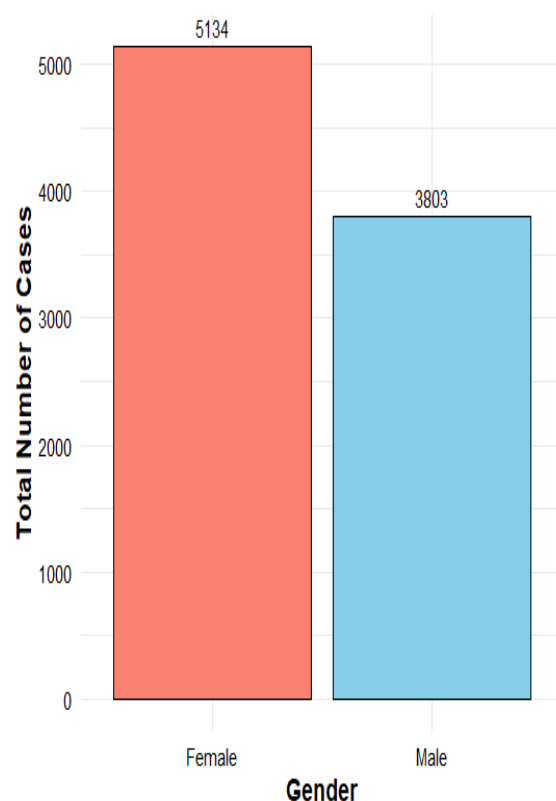


Fig. 6: The Bar-chart shows the number of malaria cases by gender and females have the highest Malaria cases across the Rural areas of Pankshin LGA, Plateau State, Nigeria.

Regarding seasonal variation (Figure 2), despite observable fluctuations with an August peak, our aggregated analysis found no statistically significant difference in mean malaria cases across months. This finding is particularly noteworthy as it appears to contradict the general understanding of malaria transmission in Nigeria, which is typically described as perennial with distinct

seasonal peaks driven by rainfall patterns.¹¹ This apparent lack of overall seasonal significance at the aggregated level might be attributed to heterogeneity among the seven clinics, where localized microclimates, diverse ecological settings, or varying intervention intensities could mask a uniform seasonal pattern when data are pooled. It could also suggest a more perennial transmission pattern in some specific rural communities within Pankshin LGA, or that the variance within monthly case counts across different years was too high to yield a statistically significant seasonal signal.

The distribution of malaria cases across age groups (Figure 3) showed highly significant differences, rejecting our null hypothesis of equal prevalence. Consistent with global epidemiological patterns, younger age groups bore a disproportionately higher burden. Specifically, the 6-16 years and 17-30 years age groups exhibited significantly higher case numbers compared to older adults (31-50 years and 51+ years). While children under five are widely recognized as highly vulnerable due to developing immune systems,^{10,3} the substantial burden in the 6-16 and 17-30 age ranges suggests that school-aged children and young adults in rural Pankshin may face increased exposure due to school attendance, play activities, or occupational demands that increase their contact with mosquito vectors during peak biting hours. Older adults likely benefit from accumulated immunity and potentially reduced exposure due to lifestyle factors.

Similarly demographic/occupational groups (Figure 4) displayed statistically significant disparities in malaria prevalence, leading to the rejection of our corresponding null hypothesis. Pregnant Women consistently showed the highest number of cases, aligning with well-established biological vulnerabilities due to altered immunity during pregnancy.^{9,12} This heightened susceptibility reveals the critical need for targeted antenatal care interventions. Students and Farmers also

bore a significantly higher burden compared to the other category. The elevated risk for farmers is consistent with literature highlighting occupational exposure due to working in agricultural settings often near mosquito breeding sites.^{2,6} The high prevalence among students may reflect their age-related vulnerability and increased exposure in communal or outdoor settings. Conversely, groups like Civil Servants and Applicants experienced lower-case numbers, likely due to reduced exposure associated with their typically indoor or less environmentally exposed occupations. This unequal distribution highlights the importance of demographic-specific approaches in malaria control.¹

The observed gender disparity (Figure 5), with females accounting for a higher total number of cases than males, is consistent with findings from other malaria-endemic regions.¹ This can be attributed to several factors, including the heightened biological vulnerability of pregnant women who are disproportionately affected by malaria,¹² as well as potential gender-specific differences in healthcare-seeking behaviors, where females may be more likely to present at clinics for diagnosis and treatment, leading to higher reported figures.

Finally, the analysis of clinic-level heterogeneity and reporting consistency (Figure 6) provided critical contextual understanding for interpreting the observed trends. The highly significant difference in mean malaria cases across clinics suggests that malaria burden is not uniform spatially within Pankshin LGA, emphasizing the need for geographically tailored interventions. Furthermore, the identified substantial gaps in reporting consistency, particularly the complete absence of data in 2018, reveal the challenges in maintaining robust surveillance systems in resource-limited rural settings.^{5,12} While the recovery in reporting from 2020 onwards is encouraging and may have contributed to the increased case numbers observed in later years by improving data capture, such inconsistencies can significantly obscure the true disease

burden and hinder effective intervention planning and resource allocation. This highlights that while control efforts are in place, challenges such as inadequate funding and limited healthcare access persist in Nigeria.^{12,2}

Recommendations

Based on the findings in rural Pankshin LGA, malaria control efforts need urgent and targeted enhancements. It's crucial to intensify existing interventions like ITN distribution and ACT access to reverse the observed increasing trend in cases, ensuring sustained funding. While overall seasonal patterns weren't statistically significant, disaggregated, clinic-specific analyses of monthly trends are vital to pinpoint local seasonal peaks for targeted interventions. Efforts must implement age-specific strategies, particularly focusing on school-aged children (6-16 years) and young adults (17-30 years) through school-based programs and broader ITN coverage. Similarly, demographic-specific control strategies are essential: strengthening antenatal care for pregnant women (IPTp/ITNs), deploying targeted vector control for farmers in agricultural settings, and reinforcing prevention programs for students. It's also important to address gender-related disparities, investigating underlying biological and behavioral factors, and ensuring equitable access to care. Finally, prioritizing interventions in identified high-burden clinics (hotspots) will optimize resource allocation. Fundamentally, enhancing and sustaining robust data reporting and surveillance systems, including digital solutions, ongoing staff training, and investigating reporting gaps, is paramount for effective, evidence-based malaria management in Pankshin.

Conclusion

This study reveals a rising trend in malaria cases across rural communities in Pankshin LGA from 2014 to 2024, with marked variations across age groups, demographics, and locations. While national

efforts have made progress, local realities such as underreported data, age-specific vulnerabilities, and hotspot communities highlight the need for more targeted, data-driven, and community-focused interventions. inconsistencies in reporting that impede accurate surveillance, emphasizing the vital role of robust and resilient health information systems. Strengthening surveillance, tailoring responses to demographic needs, and addressing high-burden areas will be critical to reducing malaria transmission and achieving long-term control in rural Nigeria.

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